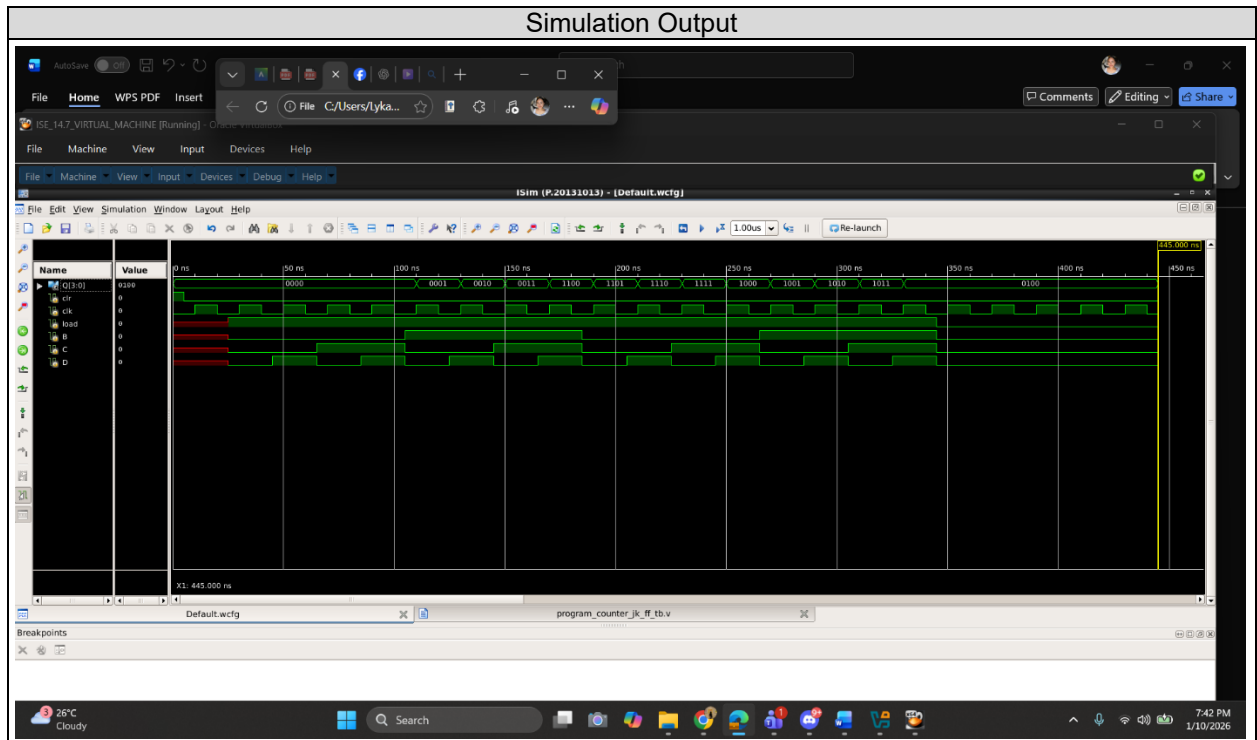


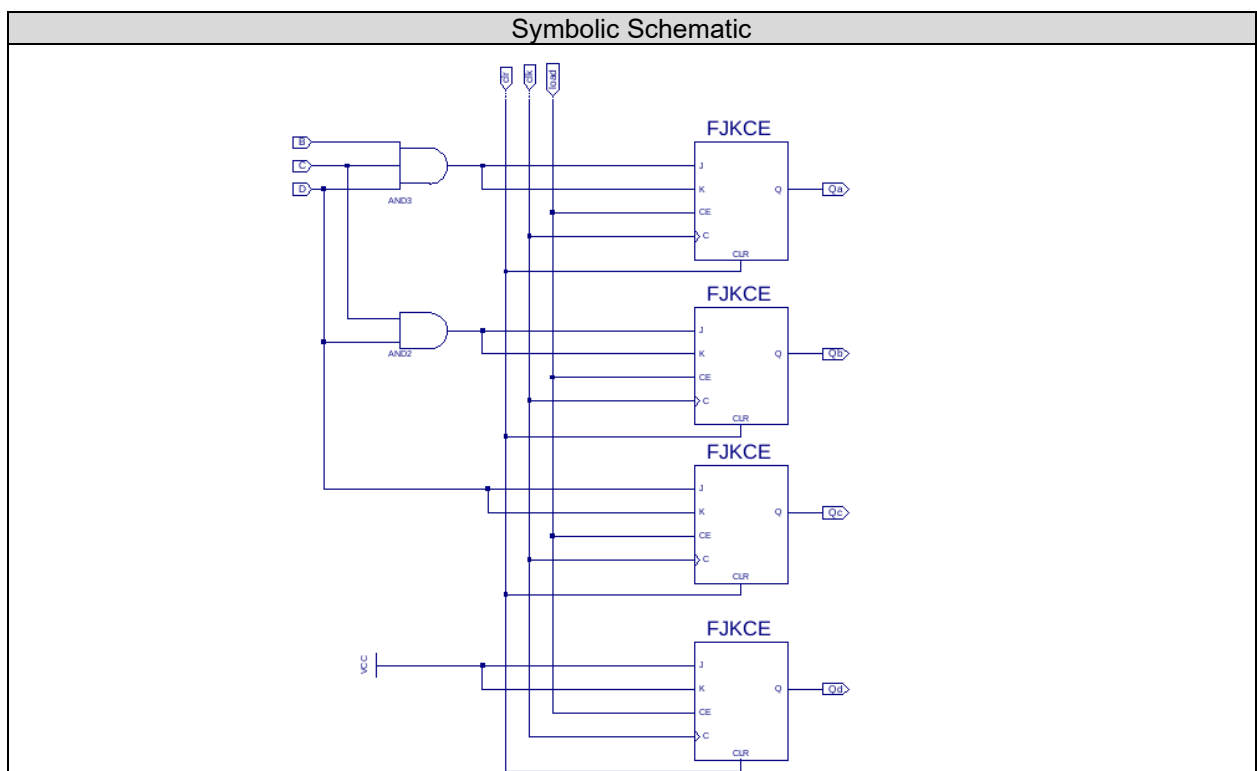
Name:	Lyka Sophia S. Jardeleza	Date:	01/10/2026
Program:	BSCpE	Section:	A461

A. DRILL 1.0 - FAMILIARIZATION OF SAP-1 COMPONENTS

1. Simulation Output

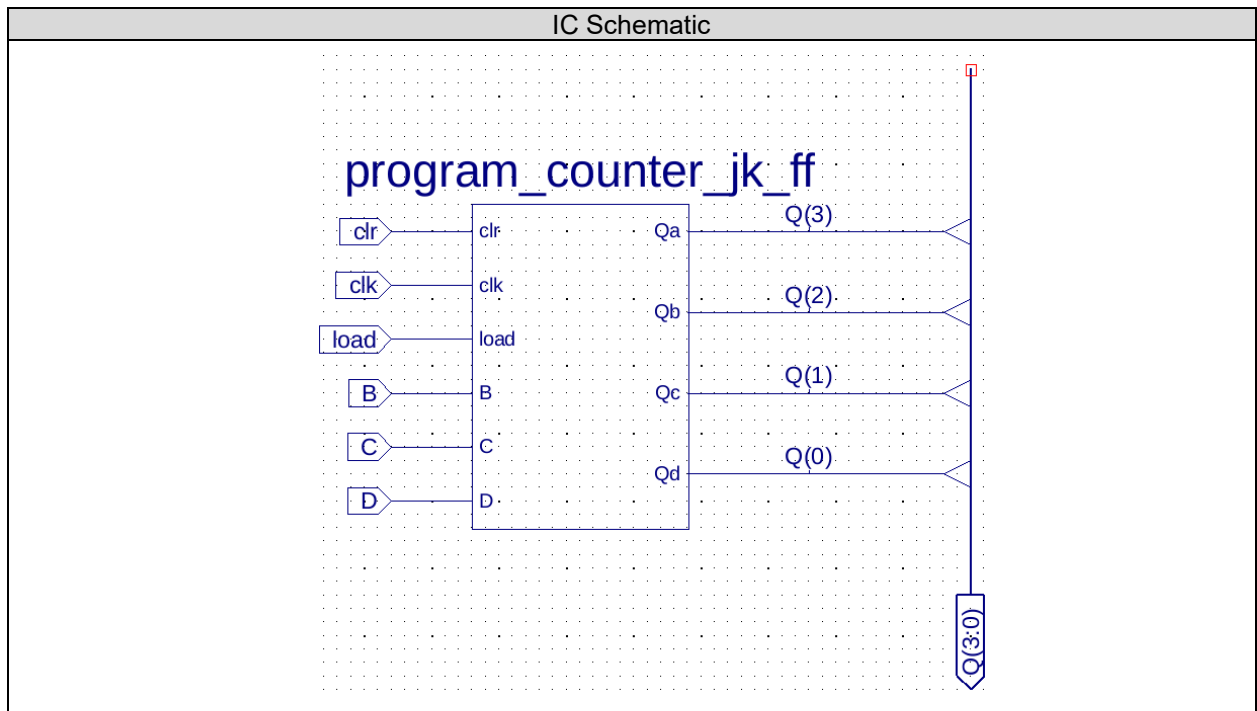


2. Symbolic Schematic

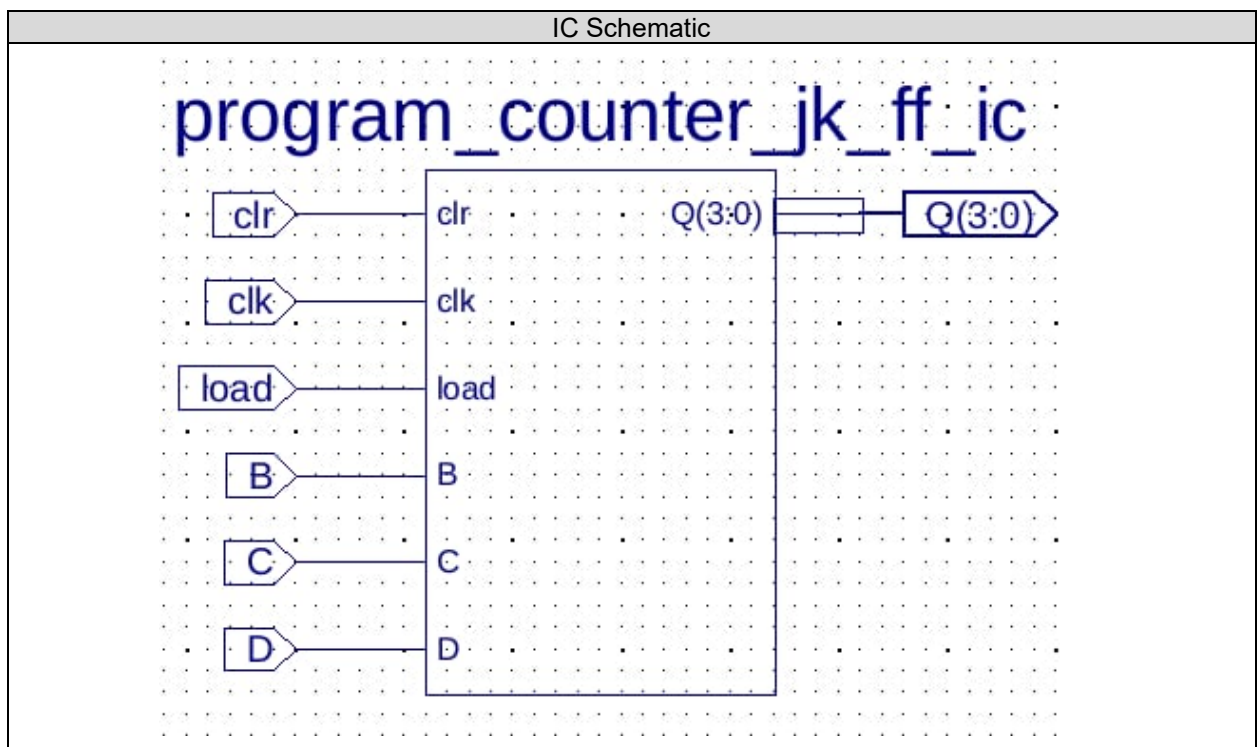




3. IC Schematic



4. Vectorized IC Schematic





4. Simulation Code

IC Schematic

```
// Verilog test fixture created from schematic
/home/ise/program_counter_jk_ff/program_counter_jk_ff_vectorized.sch - Sat Jan 10 10:46:56 2026

`timescale 1ns / 1ps

module
    program_counter_jk_ff_vectorized_program_counter_jk_ff_vectorized_sch_tb();

// Inputs
reg clr;
reg clk;
reg load;
reg B;
reg C;
reg D;

// Output
wire [3:0] Q;

// Bidirs

// Instantiate the UUT
program_counter_jk_ff_vectorized UUT (
    .clr(clr),
    .clk(clk),
    .load(load),
    .B(B),
    .C(C),
    .D(D),
    .Q(Q)
);
// Initialize Inputs
initial
    $monitor("count=%b", Q);

    initial
        begin
            #0 clr=1'b1;
            #5 clr=1'b0;

            #20 B=1'b0;C=1'b0;D=1'b0; load=1'b1;
            #20 B=1'b0;C=1'b0;D=1'b1; load=1'b1;
            #20 B=1'b0;C=1'b1;D=1'b0; load=1'b1;
            #20 B=1'b0;C=1'b1;D=1'b1; load=1'b1;

            #20 B=1'b1;C=1'b0;D=1'b0; load=1'b1;
            #20 B=1'b1;C=1'b0;D=1'b1; load=1'b1;
            #20 B=1'b1;C=1'b1;D=1'b0; load=1'b1;
            #20 B=1'b1;C=1'b1;D=1'b1; load=1'b1;

            #20 B=1'b0;C=1'b0;D=1'b0; load=1'b1;
            #20 B=1'b0;C=1'b0;D=1'b1; load=1'b1;
            #20 B=1'b0;C=1'b1;D=1'b0; load=1'b1;
            #20 B=1'b0;C=1'b1;D=1'b1; load=1'b1;
```

Laboratory Activity 1

```
initial
begin
    #20 B=1'b1;C=1'b0;D=1'b0; load=1'b1;
    #20 B=1'b1;C=1'b0;D=1'b1; load=1'b1;
    #20 B=1'b1;C=1'b1;D=1'b0; load=1'b1;
    #20 B=1'b1;C=1'b1;D=1'b1; load=1'b1;

    #20 B=1'b0;C=1'b0;D=1'b0; load=1'b0;

    #100 $stop;

    clk = 1'b0;
    forever
        #10 clk =~clk;
    end
end
endmodule
```

B. Findings ,Observations, and Comments:

Throughout the laboratory activity, I learned to design a 4-bit synchronous counter using JK flip-flops and on how to mainly use Xilinx in a virtual machine. At first, I had a hard time figuring out how to install it but eventually, I managed to solve it and made my way through it as I tried answering the activity. Also, vectorizing the schematic made the design cleaner and easier to debug and I found it impressive that a code can be generated from the schematic I made. Overall, although it was difficult at first, I had fun and it challenged me to strive harder to understand the activity and as well as the software.